

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\ a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a853bd44eda102f48.jsonl`  
- SHA-256: `c86ff61c870780077f278a9c56938f6059035cef68698f3e3b4f2869e145668b`  
- Source modified: `2026-05-31T04:17:29+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:16:19.482Z)

Web Searcher: Open VLMs for document/table extraction (broad/primary)

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Your angle: ****Open VLMs for document/table extraction (broad/primary)**** ? Primary survey of the prompt-driven VLM approach: which models, sizes, and quantizations run on a single consumer NVIDIA GPU and how they perform converting a page image to structured tables.

Search query: `best open-source vision language model 2025 document table extraction to JSON Qwen2.5-VL MiniCPM-V InternVL2.5 Phi-3.5-vision local GPU VRAM`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-05-31T04:16:22.018Z)

I'll search for the most relevant resources on this topic.

3. assistant (2026-05-31T04:16:25.053Z)

I'll search for the most relevant resources on open VLMs for document/table extraction.

4. user (2026-05-31T04:16:34.432Z)

Web search results for query: "Qwen2.5-VL document table extraction license Apache bank statement parsing local GPU VRAM requirements"

Links: [{"title": "Qwen2-VL: Hands-On Guides for Invoice Data Extraction, Video Chatting, and Multimodal RAG with PDFs", "url": "https://aihorizonforecast.substack.com/p/qwen2-vl-hands-on-guides-for-invoice"}, {"title": "Qwen3-VL-4B vs Qwen3-VL-8B: Benchmarks, VRAM Requirements, and Which to Run", "url": "https://codersera.com/blog/qwen3-vl-4b-vs-qwen3-vl-8b-benchmarks-vram-guide"}, {"title": "wanganai/qwen2.5-vl-7b-instruct · Hugging Face", "url": "https://huggingface.co/wanganai/qwen2.5-vl-7b-instruct"}, {"title": "Run Qwen2.5-VL 7B Locally: Vision AI Made Easy", "url": "https://www.labellerr.com/blog/run-qwen2-5-vl-locally/"}, {"title": "Qwen3-VL-4B: Local Vision-Language AI (Install Guide)", "url": "https://sonusahani.com/blogs/qwen3-vl"}, {"title": "Qwen2.5 VL 7B | Open Laboratory", "url": "https://openlaboratory.com/models/qwen2_5-vl-7b-instruct/"}, {"title": "Qwen3-VL-4B vs Qwen3-VL-8B: Benchmarks & VRAM Guide", "url": "https://ghost.codersera.com/blog/qwen3-vl-4b-vs-qwen3-vl-8b-benchmarks-vram-guide/"}, {"title": "Glazkov/qwen2.5-vl-table-extraction · Hugging Face", "url": "https://huggingface.co/Glazkov/qwen2.5-vl-table-extraction"}, {"title": "Qwen/Qwen2.5-VL-7B-Instruct · Hardware and vram requiremnt to run this model?", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/18"}]

Based on the search results, here's what I found regarding Qwen2.5-VL and related models for document processing:

License

Qwen2.5-VL is licensed under Apache 2.0, allowing both commercial and non-commercial use with attribution. This applies to the open-source versions of the model.

Document and Table Extraction Capabilities

Qwen2.5-VL can parse charts and tables, converting visual data into structured formats (JSON/CSV). The model supports multi-language OCR and key-value data extraction from complex backgrounds such as receipts, financial statements, invoices, and delivery bills.

Additionally, there's a specialized fine-tuned variant: a fine-tuned Qwen2.5-VL-3B model for extracting structured data from financial and economic table images, trained on synthetic table data to convert table images into JSON format with parameter, date, value, and measurement fields.

Local GPU VRAM Requirements

For the 7B model: A 7B VLM typically needs decent RAM and VRAM, often 8GB+ VRAM.

For broader context on related Qwen models with similar architectures: Qwen3-VL-4B requires a GGUF size of ~3.3 GB (Q4_K_M), with a minimum of 6 GB VRAM and comfortable operation at 8 GB. For bank statement parsing specifically, for high-volume OCR pipelines like invoice processing, 8B models with Q4_K_M at 12 GB VRAM or Q8_0 at 16 GB VRAM are recommended for better OCR accuracy on degraded or low-contrast scans.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-31T04:16:44.992Z)

Web search results for query: "best open-source vision language model 2025 document table extraction to JSON Qwen2.5-VL MiniCPM-V InternVL2.5 Phi-3.5-vision local GPU VRAM"

Links: [{"title":"Beyond Text Extraction: The 2025 Open OCR Revolution Powered by Vision-Language Models | by TechEon | Medium","url":"https://atul4u.medium.com/beyond-text-extraction-the-2025-open-ocr-revolution-powered-by-vision-language-models-89ad33d36bbf"}, {"title":"7 Best Open-Source OCR Models 2025: Benchmarks & Cost Comparison | E2E Networks","url":"https://www.e2enetworks.com/blog/complete-guide-open-source-ocr-models-2025"}, {"title":"Top 7 Open Source OCR Models - KDnuggets","url":"https://www.kdnuggets.com/top-7-open-source-ocr-models"}, {"title":"Best Open-Source Vision Language Models of 2026","url":"https://www.labelerr.com/blog/top-open-source-vision-language-models/"}, {"title":"Ultimate Guide - The Best Open Source LLM for Document Screening in 2026","url":"https://www.siliconflow.com/articles/en/best-open-source-LLM-for-Document-screening"}, {"title":"Best Vision Language Models for Document Data Extraction","url":"https://nanonets.com/blog/vision-language-model-vlm-for-data-extraction/"}, {"title":"Multimodal AI: The Best Open-Source Vision Language Models in 2026","url":"https://www.bentoml.com/blog/multimodal-ai-a-guide-to-open-source-vision-language-models"}]

Now let me search for more specific information about local GPU VRAM requirements for these models:

Links: [{"title":"GPU System Requirements Guide for Qwen LLM Models (All Variants)","url":"https://apxml.com/posts/gpu-system-requirements-qwen-models"}, {"title":"GPU System Requirement Guide for Qwen 3.5","url":"https://apxml.com/posts/qwen-3-5-system-requirement-vram-guide"}, {"title":"Best Vision Models You Can Run Locally: Every Model, Every GPU Tier | InsiderLLM","url":"https://insiderllm.com/guides/vision-models-locally/"}, {"title":"Qwen/Qwen2.5-VL-3B-Instruct · How much VRAM is required? I have 8gb RTX"}]

3060 and seems in sufficient", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/discussions/20"}, {"title": "Promptquorum", "url": "https://www.promptquorum.com/local-llms/qwen-local-deployment-guide-2026"}, {"title": "Qwen 2.5 VRAM Requirements: 9B, 27B, 35B, 72B [2026 Guide] | CraftRigs", "url": "https://craftrigs.com/guides/qwen-35-local-llm-hardware-requirements/"}, {"title": "Qwen VRAM Requirements (All Model Sizes) GIGAGPU", "url": "https://gigagpu.com/qwen-vram-requirements/"}, {"title": "System Requirements | QwenLM/Qwen2.5-VL | DeepWiki", "url": "https://deepwiki.com/QwenLM/Qwen2.5-VL/1.2-system-requirements"}, {"title": "Qwen/Qwen2.5-VL-7B-Instruct · Hardware and vram requiremnt to run this model?", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/18"}]

Let me search for more specific comparisons between MiniCPM-V and InternVL2.5:

Links: [{"title": "Benchmarking Table Extraction from Heterogeneous Scientific Extraction Documents", "url": "https://arxiv.org/pdf/2511.16134"}, {"title": "MiniCPM-Llama3-V 2.5 review: A game-changing open-source multimodal language model | by Simeon Emanuilov | Medium", "url": "https://medium.com/@simeon.emanuilov/minicpm-llama3-v-2-5-review-a-game-changing-open-source-multimodal-language-model-109d2e68989f"}, {"title": "InternVL2.5", "url": "https://internvl.github.io/blog/2024-12-05-InternVL-2.5/"}, {"title": "MiniCPM-Llama3-V 2.5 review: A game-changing open-source multimodal language model | UnfoldAI", "url": "https://unfoldai.com/minicpm-llama3-v-2-5-review/"}, {"title": "DTBench: A Synthetic Benchmark for Document-to-Table Extraction", "url": "https://arxiv.org/pdf/2602.13812"}, {"title": "Benchmarking PDF Parsers on Table Extraction with LLM-based Semantic Evaluation", "url": "https://arxiv.org/pdf/2603.18652"}, {"title": "InternVL2.5-MPO", "url": "https://internvl.github.io/blog/2024-12-20-InternVL-2.5-MPO/"}, {"title": "Evaluation of InternVL2.5 Series - InternVL's tutorials!", "url": "https://internvl.readthedocs.io/en/latest/internvl2.5/evaluation.html"}]

Based on my web search results, here's a comprehensive overview of the best open-source vision language models for 2025 focused on document table extraction to JSON with local GPU VRAM considerations:

Top Models for Document Table Extraction

Best Overall Performers

In 2025, open-source VLMs have reduced inference costs by up to 60% compared to closed commercial models, while achieving competitive benchmark scores such as MMBench >80% and MM-Vet >75% for top models like Qwen 2.5 VL and Pixtral.

For 2026, the top three recommendations are GLM-4.5V, Qwen2.5-VL-72B-Instruct, and DeepSeek-VL2?each chosen for their outstanding document understanding capabilities, multimodal reasoning, and ability to extract structured information from diverse document formats.

Specialized Models for Your Use Case

Qwen2.5-VL Series: For enterprise-scale document processing with structured data extraction from invoices, forms, and tables, Qwen2.5-VL-72B-Instruct with its 131K context window is the top choice. Model outputs include extracted tables as JSON or CSV along with beautifully formatted Markdown preserving headers and lists.

MiniCPM-V Series: MiniCPM-Llama3-V 2.5 now supports full-text OCR extraction, table-to-markdown conversion, and with 8.5 billion parameters, this model punches well above its weight, achieving performance on par with or even surpassing much larger proprietary models. It achieves 76.6% accuracy on TextVQA and 84.8% on DocVQA, setting a new bar for open-source models.

InternVL2.5: InternVL 2.5 exhibits competitive performance through extensive evaluations on a wide range of benchmarks, including document understanding and multi-image understanding.

Specialized OCR Models: dots.ocr is an open-source 1.7B parameter vision-language transformer model that combines layout detection and content recognition, and released in July 2025 under MIT license, achieves state-of-the-art performance for text, tables on OmniDocBench.

Local GPU VRAM Requirements

****Qwen2.5-VL 7B****: Qwen2-VL 7B runs in ~6 GB VRAM via `ollama pull qwen2-vl:7b` and reads Chinese, Japanese, and Korean text at up to 4096x4096 resolution.

****Phi-3.5-Vision****: Phi-4-reasoning-vision requires ~9-10 GB at Q4_K_M and fits on a 12 GB GPU comfortably.

****General Hardware Guidance****: For hardware decisions: 12 GB VRAM ? 14B model; 24 GB VRAM ? 32B; 48 GB+ (two GPUs or Apple Silicon 64 GB) ? 72B.

****Document OCR Focus****: PaddleOCR-VL 0.9B (specialized, 109 languages) is better when you need exact layout preservation ? tables, formulas, column structures, while Qwen models are better when you need the model to understand the document and answer questions about its content.

The choice between these models depends on your specific hardware constraints and whether you prioritize exact table structure preservation or general document understanding with reasoning capabilities.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:16:47.852Z)

I have strong results. Let me run two more targeted searches to nail down the weights-license nuances (per-size Qwen licensing, GOT-OCR2.0, olmOCR, dots.ocr) and hallucination/reliability on financial numbers.

7. user (2026-05-31T04:16:56.771Z)

Web search results for query: "Qwen2.5-VL 3B weights license non-commercial research vs 7B 72B Apache InternVL2.5 license MiniCPM-V model weights commercial"

Links: [{"title": "Qwen Promises AI Agent Capabilities From It Free-to-Use Qwen2.5-VL Vision Language Models - Hackster.io", "url": "https://www.hackster.io/news/qwen-promises-ai-agent-capabilities-from-it-free-to-use-qwen2-5-vl-vision-language-models-db51344a23b4"}, {"title": "LICENSE · Qwen/Qwen2.5-VL-72B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE"}, {"title": "Alibaba Debuts Qwen2.5-VL, A Powerful Family of Open Vision-Language Models", "url": "https://www.deeplearning.ai/the-batch/alibaba-debuts-qwen2-5-vl-a-powerful-family-of-open-vision-language-models/"}, {"title": "LICENSE · Qwen/Qwen2.5-VL-3B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE"}, {"title": "qwen2.5vl/license", "url": "https://ollama.com/library/qwen2.5vl:latest/blobs/832dd9e00a68"}, {"title": "LICENSE · Qwen/Qwen2.5-72B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-72B-Instruct/blob/main/LICENSE"}, {"title": "wangkanai/qwen2.5-vl-7b-instruct · Hugging Face", "url": "https://huggingface.co/wangkanai/qwen2.5-vl-7b-instruct"}, {"title": "nvidia/Qwen2.5-VL-7B-Instruct-FP8 · Hugging Face", "url": "https://huggingface.co/nvidia/Qwen2.5-VL-7B-Instruct-FP8"}, {"title": "Qwen/Qwen2.5-VL-7B-Instruct · Qwen2.5-VL-7B-Instruct License?", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/31"}]

Links: [{"title": "About the license of internVL2.5-4B model · Issue #900 · OpenGVLab/InternVL", "url": "https://github.com/opengvlab/internvl/issues/900"}, {"title": "OpenGVLab/InternVL2_5-2B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL2_5-2B"}, {"title": "OpenGVLab/InternVL2_5-Pretrain-Models · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL2_5-Pretrain-Models"}, {"title": "Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling", "url": "https://arxiv.org/pdf/2412.05271"}, {"title": "OpenGVLab/InternVL3_5-8B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3_5-8B"}, {"title": "Alibaba Debuts Qwen2.5-VL, A Powerful Family of Open Vision-Language Models", "url": "https://charonhub.deeplearning.ai/alibaba-debuts-qwen2-5-vl-a-powerful-family-of-open-vision-language-models/"}, {"title": "InternVL2.5", "url": "https://internvl.github.io/blog/2024-12-05-InternVL-2.5/"}, {"title": "Introduction of InternVL2.5 Series - InternVL's tutorials!", "url": "https://internvl.readthedocs.io/en/latest/internvl2.5/introduction.html"}]

Links: [{"title": "minicpm-v/license", "url": "https://ollama.com/library/minicpm-"}]

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v:latest/blobs/8603ca877636"}, {"title": "GitHub - OpenBMB/MiniCPM-V: A Pocket-Sized MLLM for Ultra-Efficient Image and Video Understanding on Your Phone · GitHub", "url": "https://github.com/OpenBMB/MiniCPM-V"}, {"title": "MiniCPM-V 4.6 1.3B - Intelligence, Performance & Price Analysis", "url": "https://artificialanalysis.ai/models/minicpm-v4-6-1-3b"}, {"title": "minicpm-v:8b-2.6-q4_0/license", "url": "https://ollama.com/library/minicpm-v:8b-2.6-q4_0/blobs/8603ca877636"}, {"title": "MiniCPM-V 4.6 1.3B - Dataconomy", "url": "https://dataconomy.com/ai-models/minicpm-v4-6-1-3b/"}, {"title": "minicpm-v:8b/license", "url": "https://ollama.com/library/minicpm-v:8b/blobs/8603ca877636"}, {"title": "openbmb/MiniCPM-V · Hugging Face", "url": "https://huggingface.co/openbmb/MiniCPM-V"}, {"title": "minicpm-v:8b-2.6-q4_K_M/license", "url": "https://ollama.com/library/minicpm-v:8b-2.6-q4_K_M/blobs/8603ca877636"}, {"title": "minicpm-v:8b-2.6-q3_K_S/license", "url": "https://www.ollama.com/library/minicpm-v:8b-2.6-q3_K_S/blobs/8603ca877636"}, {"title": "minicpm-v:8b-2.6-q4_1/license", "url": "https://ollama.com/library/minicpm-v:8b-2.6-q4_1/blobs/8603ca877636"}]
```

Based on the search results, here's a comprehensive summary of the licensing information for these vision-language models:

Qwen2.5-VL

The Qwen2.5-VL weights are available under different licenses depending on the model size: Qwen2.5-VL-3B is free for non-commercial uses, Qwen2.5-VL-7B is free for commercial and noncommercial uses under the Apache 2.0 license, and Qwen2.5-VL-72B is free to developers that have less than 100 million monthly active users.

InternVL2.5

InternVL2.5 is released under the MIT License. Unlike Qwen, this provides a consistent, permissive license across all model sizes that allows both commercial and non-commercial use. The project uses the pre-trained internlm2_5-1.8b-chat as a component, which is licensed under the Apache License 2.0.

MiniCPM-V

The MiniCPM-o/V model weights and code are open-sourced under the Apache-2.0 license. However, there are usage restrictions: If services or products deploy the model on edge-side devices not exceeding 5,000 units, or provide applications with a daily active user count (DAU) of less than 1 million, users can apply to OpenBMB for permission and, after filling out a registration questionnaire, may be allowed to use it commercially for free. Otherwise, please email to apply for authorization from OpenBMB, which may, at its discretion, grant permission.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-05-31T04:16:57.874Z)

Web search results for query: "VLM OCR hallucination financial numbers dropped rows reliability GOT-OCR2.0 olmOCR dots.ocr benchmark OmniDocBench table extraction"

Links: [{"title": "olmOCR 2 Unit Test Rewards for Document OCR Jake Poznanski Luca Soldaini", "url": "https://arxiv.org/pdf/2510.19817"}, {"title": "1 Performance of GLM-OCR on OmniDocBench v1.5.", "url": "https://arxiv.org/html/2603.10910v1"}, {"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "OCR Model Leaderboard 2026 - Benchmarks and Which to ...", "url": "https://instavar.com/blog/ai-production-stack/OCR_SOTA_Feb_2026_Open_Document_AI_Leaderboard"}, {"title": "DeepSeek-OCR-OmniDocBench/olmocr/README.md at main · alphaXiv/DeepSeek-OCR-OmniDocBench", "url": "https://github.com/alphaXiv/DeepSeek-OCR-OmniDocBench/blob/main/olmocr/README.md"}, {"title": "olmOCR 2 Unit Test Rewards for Document OCR", "url": "https://arxiv.org/html/2510.19817v1"}, {"title": "OlmOCR-Bench Review: Insights and Pitfalls | LlamaIndex", "url": "https://www.llamaindex.ai/blog/olmocr-bench-review-insights-and-pitfalls-on-an-ocr-benchmark"}, {"title": "FireRed-OCR Technical

Report", "url": "https://arxiv.org/pdf/2603.01840"}]]

I found relevant search results about VLM OCR benchmarking and related topics. Here's what the search revealed:

Key OCR Models and Benchmarks

Recent VLM-based OCR systems include end-to-end models like Nanonets-OCR2-3B, MinerU 2.5, dots.OCR, Monkey OCR, and Chandra OCR, as well as hybrid systems like DeepSeek-OCR and PaddleOCR-VL.

OmniDocBench and Performance Comparisons

On OmniDocBench v1.5, GLM-OCR achieved the highest overall score of 94.6, followed by PaddleOCR-VL-1.5 at 94.5, while dots.ocr scored 88.4. FireRed-OCR ranks first in the overall metric and demonstrates leading performance across all sub-categories including text recognition, formula processing, table structure analysis, and reading order inference.

OlmoOCR-Bench Benchmark

OlmoOCR-Bench is a unit-test-driven, end-to-end evaluation benchmark designed to assess the practical correctness of OCR systems on diverse, challenging real-world documents, specifically tailored for benchmarking VLMs tasked with extracting clean, structured text including mathematical formulas, tables, multi-column layouts, and scanned material.

Common OCR Failure Modes

Common failure modes include residual instances of repeated n-grams or hallucinated text in long-tiny-text settings. Large Vision-Language Models frequently suffer from "structural hallucination" when processing complex documents, limiting their utility in industrial OCR applications.

Table Extraction Improvements

olmoOCR 2 shows the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:17:04.102Z)

I now have comprehensive coverage across the primary VLM survey, VRAM tiers, the critical per-size license nuances, and hallucination/reliability on financial tables. I have what I need to rank the results.

Key findings to highlight in my selection:

- The deeplearning.ai "The Batch" piece nails the critical per-size Qwen2.5-VL licensing distinction (3B = non-commercial, 7B = Apache-2.0, 72B = restricted by MAU), which is load-bearing for muneem's commercial-safety constraint.
- The Nanonets blog and BentoML guide are strong primary surveys of VLMs for document/table-to-JSON extraction.
- APXML / InsiderLLM give concrete VRAM tiers per model size.
- The OmniDocBench leaderboard and olmoOCR-bench cover the reliability/hallucination angle directly.

Let me compile the ranked shortlist.

10. user (2026-05-31T04:17:22.884Z)

Structured output provided successfully

11. assistant (2026-05-31T04:17:29.283Z)

I've completed the search and returned the ranked shortlist via StructuredOutput. The six results cover the primary VLM survey angle plus the load-bearing license and reliability dimensions for muneem:

1. ****deeplearning.ai / The Batch**** ? the critical per-size Qwen2.5-VL weights licensing (3B = non-commercial, 7B = Apache-2.0, 72B = MAU-restricted). This is the single most decision-relevant source given muneem's commercial-safety rule.
2. ****Nanonets**** ? primary survey of prompt-driven VLMs for document/table-to-JSON.
3. ****BentoML**** ? self-hosting-oriented VLM survey (vLLM on single GPU).
4. ****InsiderLLM**** ? concrete VRAM-tier-to-model/quantization mapping.
5. ****Instavar OCR Leaderboard 2026**** ? OmniDocBench table-structure scores for OCR-specialist VLMs, surfaces MIT-licensed dots.ocr.
6. ****olmOCR 2 (arXiv)**** ? VLM hallucination failure modes on numbers/tables, reinforcing the balance-reconciliation verification approach.